Pawar Lab

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Holobiont Microbiomes: Bioinformatic Analysis of Insect Microbial Communities

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Research Objectives

- Comparing the **gut** microbial **composition and diversity** of insects of varying **sizes** and analyzing the influence of these microbes on the insects' metabolic performance.
- Data resource: https://www.ncbi.nlm.nih.gov
- Key tool: **16S rRNA Sequencing Analysis using QIIME2**

Research Questions

1. How does insect size affect the composition and diversity of their gut microbiome?

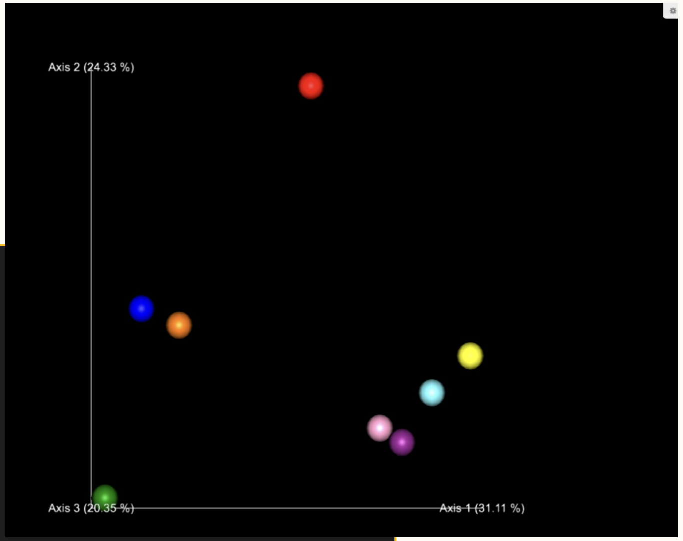
2. What is the relationship between the gut microbial diversity of insects of different sizes and their metabolic performance?

Metabolic rate vs diversity / Metabolic rate vs insect size

Preliminary Results

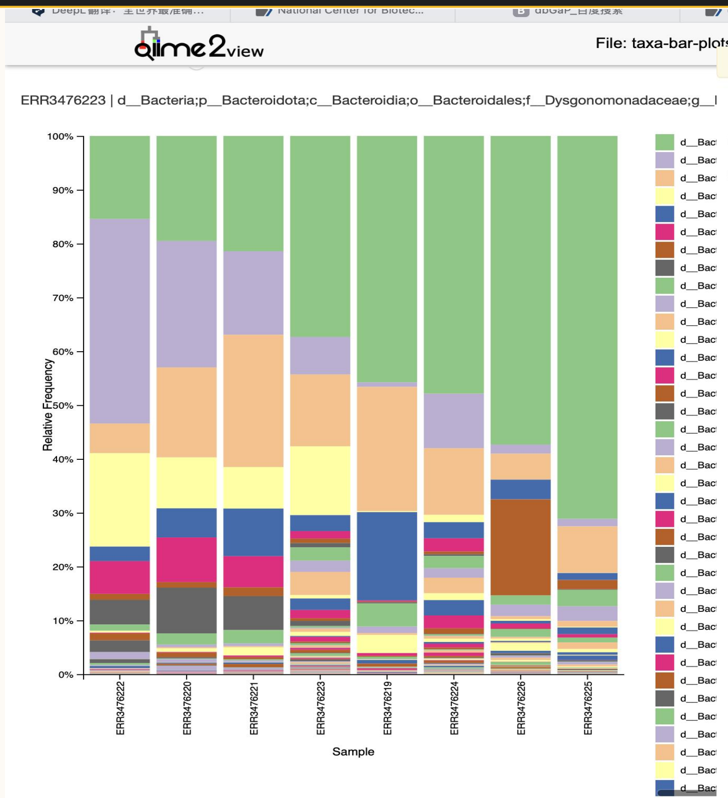
Pipeline Steps:

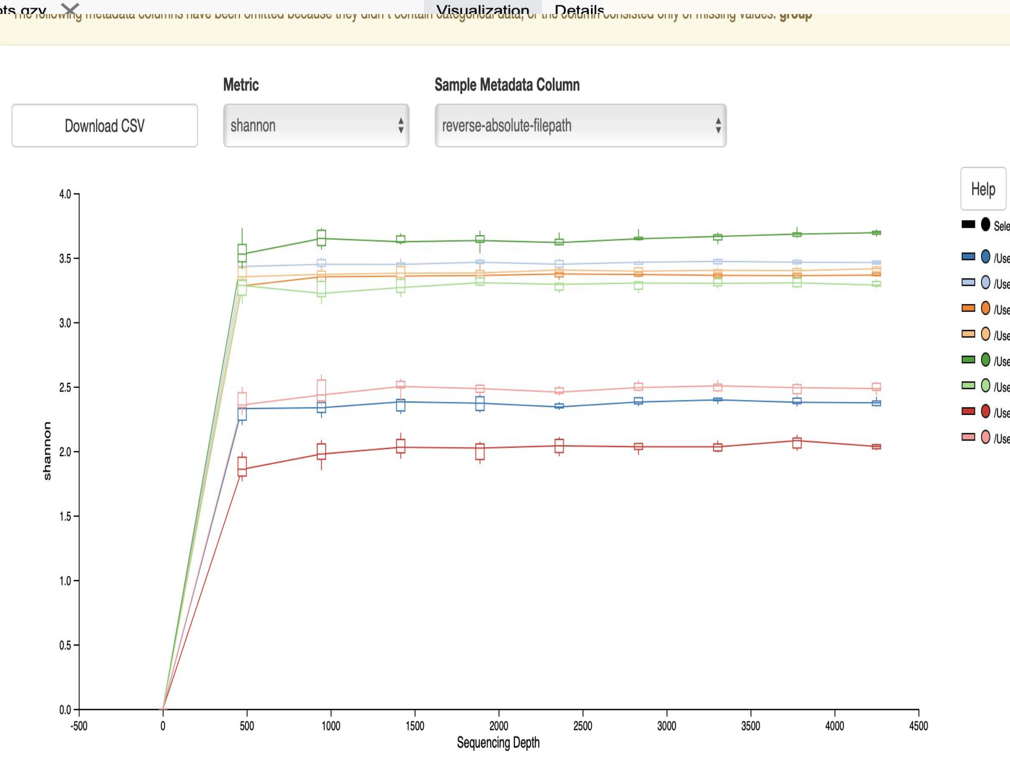
1. Preparation for data download.
2. Downloading SRA data relevant data regarding insect gut microbiomes from the NCBI website.
3. Converting SRA data to FASTQ format.
4. Sequencing data quality control and processing with FastQC software.
5. 16S rRNA gene sequencing analysis using QIIME.



xiepipeline:

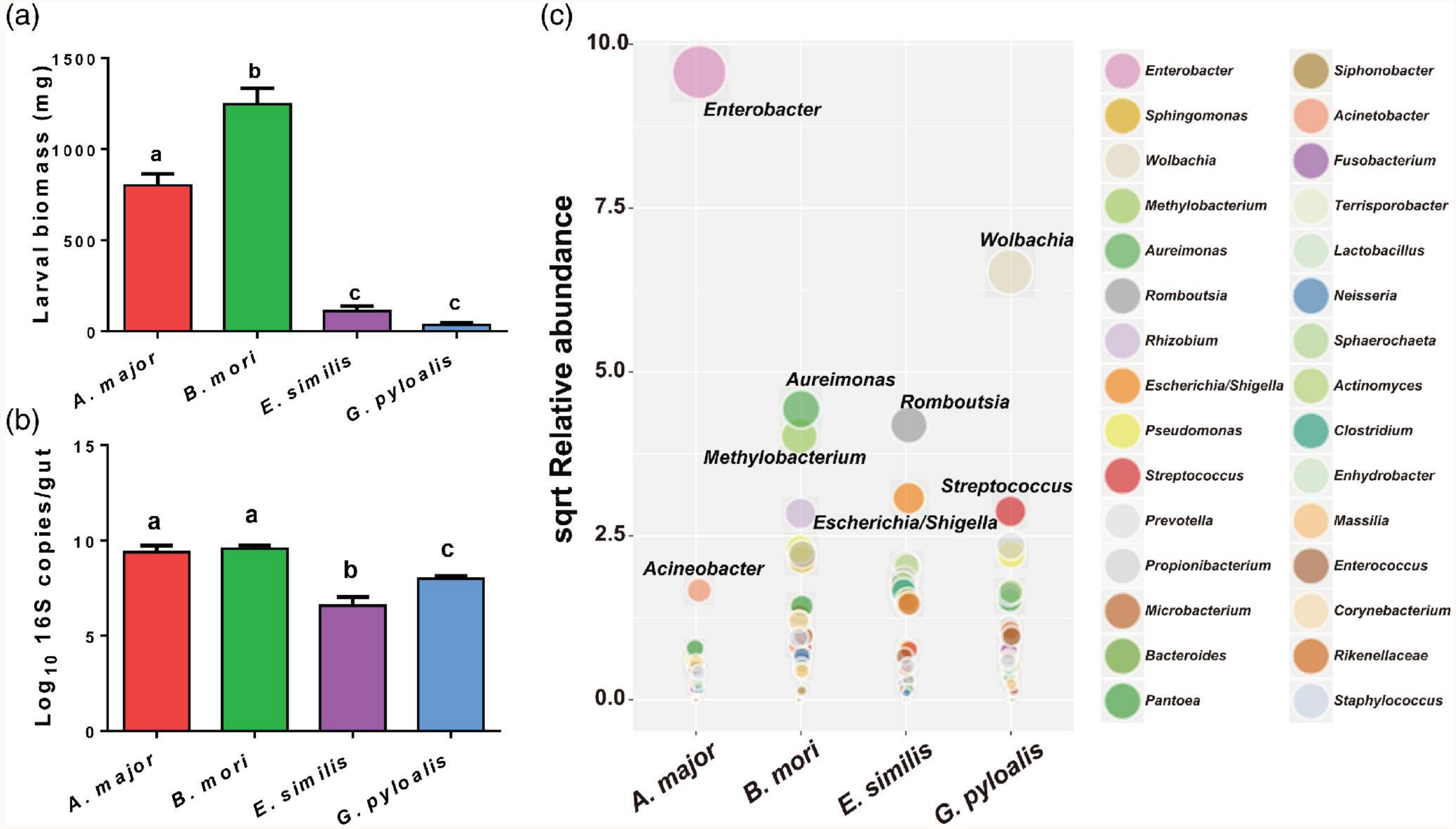
01. Trim data using Cutadapt
02. Denoise data using DADA2
03. Remove chimeras using Vsearch
04. OTU clustering with vsearch and OTU species classification with silva/greengenes
05. Phylogenetic tree construction
06. Species taxonomic annotation and generation of OTU table
07. Diversity and composition analysis





Expected Results

Hypothesis 1：The larger the body size of an insect, the richer the microbial diversity and more complex the composition of its gut microbiota



Expected Results

Hypothesis 2: Insects' overall metabolic rate increases with body size, but metabolic rate per unit body weight may decrease

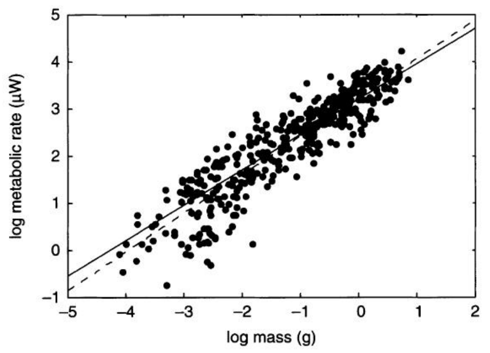
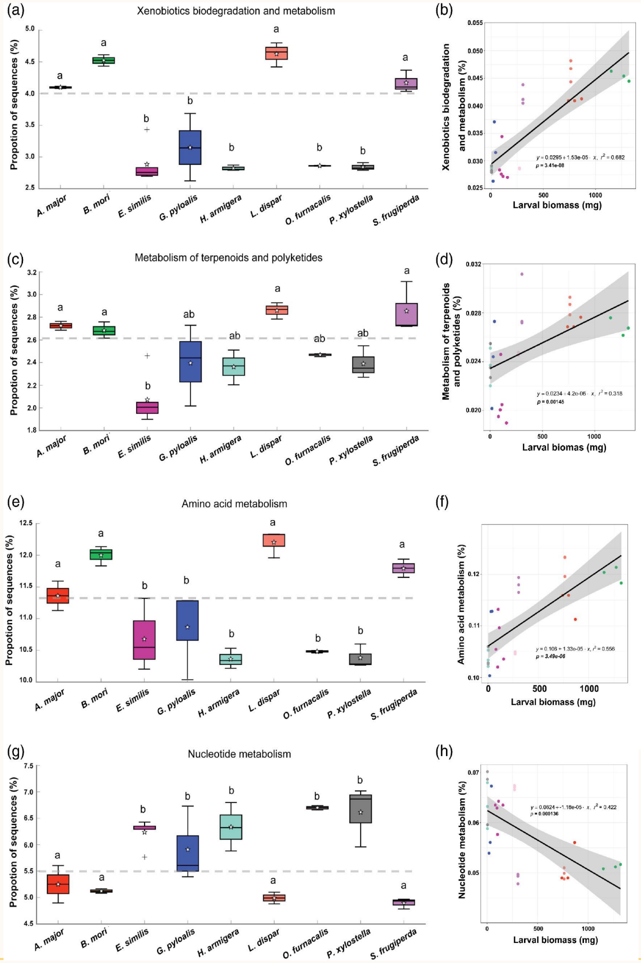


Fig. 2. Interspecific relationship between $\log_{10}$ mass (g) and $\log_{10}$ metabolic rate (μW) in insects. The data are adjusted to 25 °C (see Materials and methods) and are for 391 species from 16 orders. Ordinary least squares relationship (stippled line) described by the equation $\log_{10}$ metabolic rate = 3·26 + 0·82 × $\log_{10}$ mass. 95% confidence intervals for the mass exponent: 0·78 – 0·85. Relationship based on the phylogenetically independent contrasts (solid line), described by the equation $\log_{10}$ metabolic rate = 3·20 + 0·75 × $\log_{10}$ mass. Ninety-five per cent confidence intervals for the mass exponent: 0·70 – 0·79.



PREVIOUS ARTICLE NEXT ARTICLE

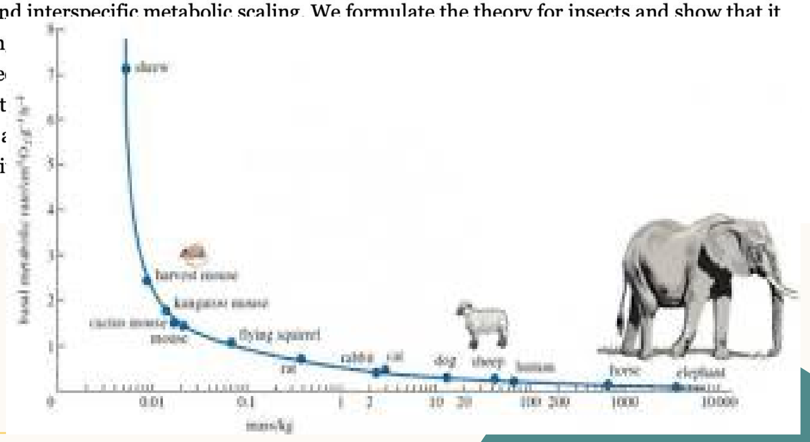
Ontogenetic and Interspecific Metabolic Scaling in Insects

James L. Maino, Michael R. Kearney, Associate Editor: Tony D. Williams, and Editor: Judith L. Bronstein

PDF PDF PLUS Abstract Full Text Supplemental Material

Abstract

Design constraints imposed by increasing size cause metabolic rate in animals to increase more slowly than mass. This ubiquitous biological phenomenon is referred to as metabolic scaling. However, mechanistic explanations for interspecific metabolic scaling do not apply to ontogenetic size changes within a species, implying different mechanisms for scaling phenomena. Here, we show that the dynamic energy budget theory approach of compartmentalizing biomass into reserve and structural components provides a unified framework for understanding ontogenetic and interspecific metabolic scaling. We formulate the theory for insects and show that it can account for ontogenetic metabolic scalin during pupation. After correcting for the pre of respiration between species is approximat scaling and our theoretical predictions. The a consistent framework suggests that the parti metabolic theory.

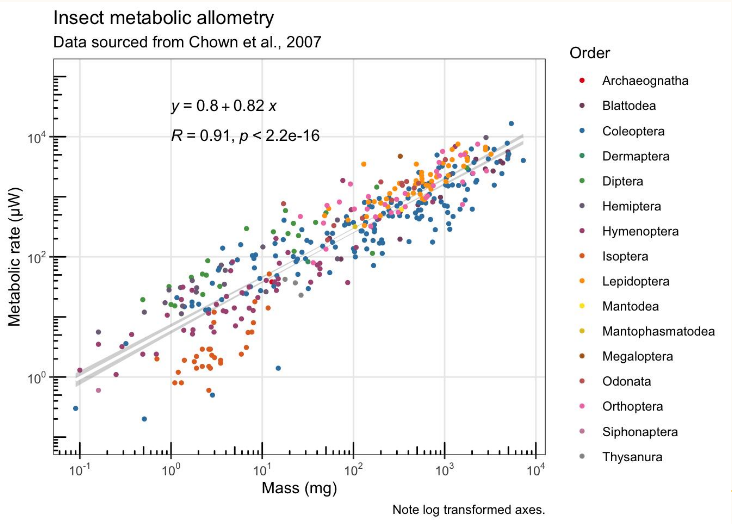


Aspects of the Holobionts under Climate change problem

Microbiome Diversity

Temperature

Insect size



Metabolic Rate

THANK YOU

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